Optimized GNSS Network Station Selection to Support the Development of Ionospheric Threat Models for GBAS

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ABSTRACT

Extremely large ionospheric spatial gradients present potential integrity threats to the users of global navigation satellite systems (GNSS) augmentation systems. Thus, these ionospheric anomalies need to be monitored by ground reference stations and users must be alarmed within time-to-alerts. The long-term ionospheric anomaly monitoring (LTIAM) tool has been developed to monitor ionospheric behavior continuously over the life cycle of ground-based augmentation systems (GBAS) and to build ionosphere threat models for all regions where GBAS will be fielded in the future. However, the use of poor-quality GNSS data degrades the accuracy of ionospheric delay estimates, produces many faulty anomaly candidates and thus adds a great burden to LTIAM processing.

This paper develops an optimized method to select well-distributed, high-quality stations for a GNSS network. Thresholds which maximize the elimination of spurious gradients while minimizing unnecessary station removals are established. This method achieved an 89% reduction of faulty anomaly candidates while discarding only 16% of the more than 1500 CORS stations. Because this method is also used to remove redundant stations in dense regions with a desired baseline constraint, the benefit when using it is that it reduces the processing time of LTIAM. These algorithms are also tested using data collected from other GNSS reference station networks to verify the performance of the proposed method.

1.0. INTRODUCTION

Single-frequency-based GNSS augmentation systems require the development of ionospheric threat models to insure that users are sufficiently protected against ionospheric anomalies. The Long-Term Ionospheric Anomaly Monitoring (LTIAM) tool is an automated software package developed to build ionospheric anomaly threat models from observed events by analyzing past data and continuously monitoring ionospheric behavior over the life cycle of Ground-Based Augmentation Systems (GBAS) [1 – 4]. The existing ionospheric anomaly threat model for the Conterminous U.S. (CONUS) is based on analysis of the past ionospheric storms which occurred during the last solar peak in 2000-2004 [5, 6]. As the current solar cycle approaches its maximum in 2013-2014, continuous monitoring of ionosphere becomes of particular importance to confirm the validity of this threat model and update it if it is not sufficient to bound newly-discovered events.

The LTIAM tool automatically processes data collected from today's denser Continuously Operating Reference Stations (CORS) network in CONUS, which has over 1500 stations as of 2012 compared to about 400 stations prior to 2004. As the total number of stations increases, the observability of ionospheric behavior improves, but the number of stations with poor GPS data quality also increases. Thus, while automated measurement screening is included in LTIAM, the use of poor-quality data produces many cases of faulty anomaly candidates and burdens the process significantly.

Previous work developed a comprehensive means of determining the quality of GPS data to identify stations that are poor according to multiple quality metrics (or "parameters")_[7]. The use of this method on the CORS network in CONUS eliminated about 87% of false gradients. However, thresholds to remove poor stations were established independently for each quality metric without considering the interrelations between other quality metrics. Because the violation of any one threshold results in exclusion, this method eliminates more stations than necessary.

This paper develops a new station-selection algorithm to establish thresholds which maximize the elimination of spurious gradients while minimizing unnecessary station removals. In this algorithm, a set of thresholds for the quality parameters is determined by solving this optimization problem in a two-dimensional (i.e., station and quality parameter) iterative way. This method first searches for the most "desired to be removed" station in a sense that false gradients are maximally eliminated while good-quality stations are minimally removed. At the same time the most "efficient" quality parameter is selected in a manner that unnecessary removal of other stations is avoided as much as possible from setting a threshold of this parameter to remove the most "desired to be removed" station. The threshold of the selected parameter only is lowered at each step to discard poor-quality stations, and this process is iterated until the desired percentages of faulty candidates are removed.

Regardless of data quality, a large number of stations bring an excessive computation load to the automated post-processing of data by LTIAM. As long as an area of interest is covered by a sufficient number of stations with separations of less than a desired baseline distance (as an example, 40 km), other nearby stations in the region which do not improve the geographic observability of the ionospheric behavior are redundant and can be removed even if they provide high-quality data. This paper therefore proposes a method to select a subset of stations passing the data-quality method based upon their geographical distribution. The sub-network is composed of the minimal set of stations with which an area originally covered by stations with a desired baseline distance is still maintained.

The optimized method to select well-distributed, high-quality stations, when implemented, allows a dramatic reduction of both an automated processing time and a manual analysis burden in LTIAM,.and consequently help to return more reliable LTIAM results. In this paper, Section 2.0 illustrates GNSS reference stations with poor-quality data and the problem of using those data for ionosphere analysis. Section 3.0 explains the optimized GNSS network station-selection methodology in detail. Section 4.0 and Section 5.0 show the results of applying this method to CORS stations in CONUS and other GNSS reference station networks, respectively. Section 6.0 concludes the paper.

2.0. POOR CORS DATA QUALITY

This section investigates the effect of GPS data collected from stations with poor data quality on ionospheric delay and gradient estimation. Figure 1 shows some real examples of poorly sited CORS stations in CONUS. The data from these stations which are located next to a solar panel, in a bush, and even in between towers may produce erroneous estimates of ionospheric delays, and consequently unusually large ionospheric gradients due to corrupted raw measurements.

Figure 1. Real Eaxmples of Poorly Sited CORS Stations in CONUS [8]

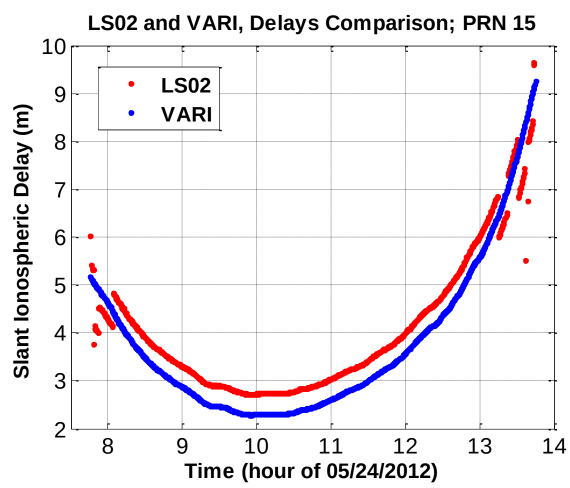


Figure 2. Dual-frequency Slant Ionospheric Delay Estimates for Stations LS02 (Poor Quality Data) and VARI (Good Quality Data)

Figure 22 shows the slant ionospheric delays observed from two nearby stations LS02 and VARI while they tracked PRN 15 on 24 May 2012. LS02 is a good example of a station with poor GPS data quality. From the many fragments of ionospheric delay estimates from LS02 (red), it is evident that its carrier-phase measurements are corrupted by numerous cycle slips, resulting in outliers and short arcs of ionospheric observables. Figure 33 shows the estimated ionospheric spatial gradients between LS02 and VARI. The ionospheric-delay leveling errors due to the short arcs and the large gradients due to the excessive outliers from LS02 are observable at each end of the curve. This example illustrates how poor data quality degrades the accuracy of ionospheric delay estimation and can produce extremely large ionospheric gradients that are not real.

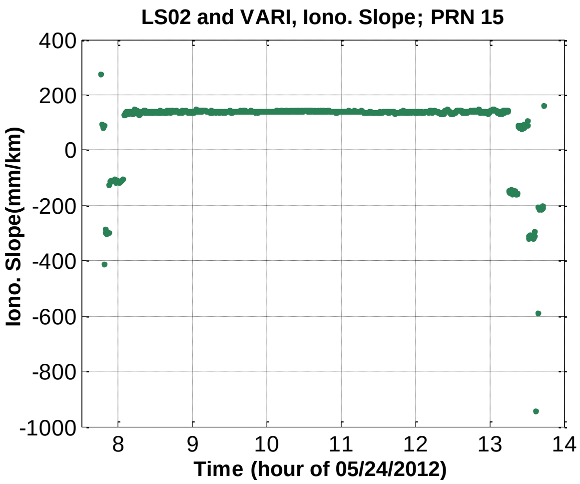


Figure 3. Ionospheric Gradient Estimates Corrupted by Poor Quality Data from Station LS02

3.0. DATA ANALYSIS METHODOLOGY

The methodology for optimally selecting well-distributed, high-quality stations within a GNSS reference station network is composed of three steps: measuring GNSS data quality information using several metrics, determining the thresholds of each of these metrics to remove stations with poor-quality data, and re-examining not only tentatively excluded poor-quality stations but also redundant good-quality stations in a dense area considering their geographical locations. First, data quality information is obtained by processing the RINEX file collected at each station through a series of data-quality-measurement algorithms [7]. Second, stations with sufficiently high-quality data are selected based on whether the quality parameters of the station exceed established thresholds. Third, to observe anomalous ionospheric gradients throughout an area of interests, the selected stations should cover the area with separations of less than 40 km to the degree possible. To meet this criterion in regions with relatively few stations, some degrees of data quality may need to be sacrificed. Therefore, we examine the geographical contribution of stations which have not been selected due to poor data quality. Stations whose location increases the geographical observability of ionospheric behavior are restored despite their poor data quality. On the other hand, stations with good data quality may be geographically redundant if located in a dense area. To select a subset of stations which are geographically necessary among good-quality stations, stations whose location does not improve the observablility of ionosphere and only increase a computational load are removed despite their good data quality. The details of each step are described in the following subsections.

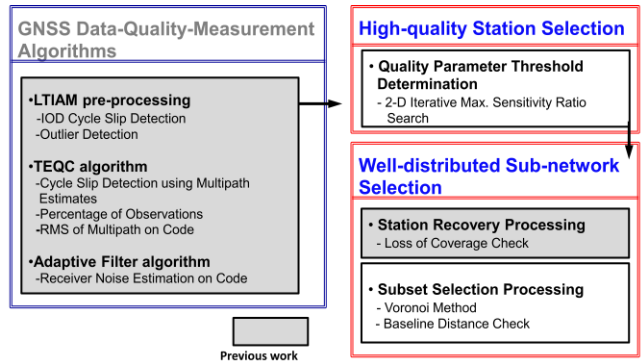


Figure 4. Optimized GNSS Network Station Selection Algorithms

3.1. DATA QUALITY MEASUREMENT ALGORITHMS

The GNSS data-quality measurement algorithms have been developed and described in [7]. The input of these algorithms is the RINEX file collected from a station of interest, and the output is the GNSS data quality information of the corresponding station. These algorithms are composed of mainly three parts: LTIAM pre-processing, the "Translation, Editing, and Quality Check (TEQC)" algorithm, and the adaptive filter algorithm. In the LTIAM pre-processing step, the number of IOnospheric Delay (IOD) cycle slips, the number of outliers, and the number of short arcs are counted as separate data-quality measurements. Second, we implemented the TEQC quality metrics developed by the University Navstar Consortium (UNAVCO), which are commonly used to solve pre-processing problems [9, 10]. The TEQC method is used to obtain quality information, which includes the percentage of observations, the mean of multipath on L1 code and L2 code, and the number of cycle slips detected using multipath estimates. Third, an adaptive filter algorithm is designed to estimate receiver noise on code measurements (see [7] for details).

3.2. HIGH QUALITY STATION SELECTION

Among the GPS data quality metrics, seven quality parameters have the greatest impact on LTIAM performance. Those are the number of IOD cycle slips, number of short arcs, number of outliers, percentage of (valid) observations, multipath on L1 code and L2 code, and latency. 'Latency' indicates the number of days for which the RINEX files are properly loaded. Figures 5 and

6 show the procedure of determining the thresholds of these seven quality parameters and selecting high-quality stations. After obtaining data quality measurements for all CORS stations in CONUS, we rank stations in order from the worst to the best for each quality parameter. The worst station is on the top and the best station is in the bottom in Figure 6. The black line shows the hypothetical location of the threshold for each parameter (any stations above the threshold in this figure are to be removed). The blue boxs and red boxs represent normal stations and troublesome stations which generate faulty anomaly candidates, respectively.

As the threshold is reduced, both the number of stations removed, $M_{stations}$, and the number of faulty ionospheric anomaly candidates removed, $N_{candidates}$, increase. A troublesome station which generates the largest number of faulty candidates needs to be removed first. However, the removal of too many stations with acceptable data quality in order to remove the particular station also needs to be avoided. To maximize the elimination of false gradients while minimizing good-quality station removals, we measure the sensitivity ratio, λ, that represents the ratio of the number of faulty ionospheric gradients removed to the number of stations removed when the threshold is lowered to the ith station:

$$\lambda = \frac{N_{candidates}(i)}{M_{stations}(i)}, \tag{1}$$

where i is the rank of stations ($i = 1, 2, ..., n$).

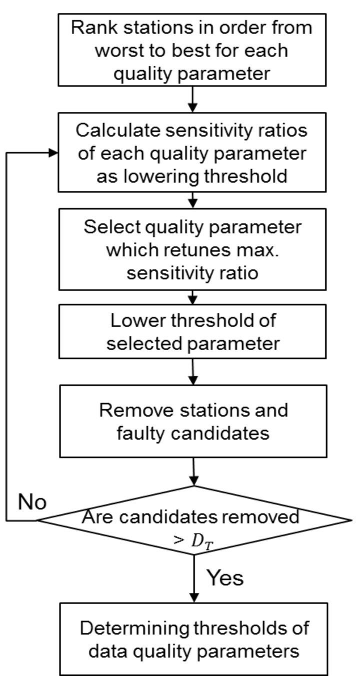


Figure 5. Procedure of High-Quality Station Selection Alogrithm

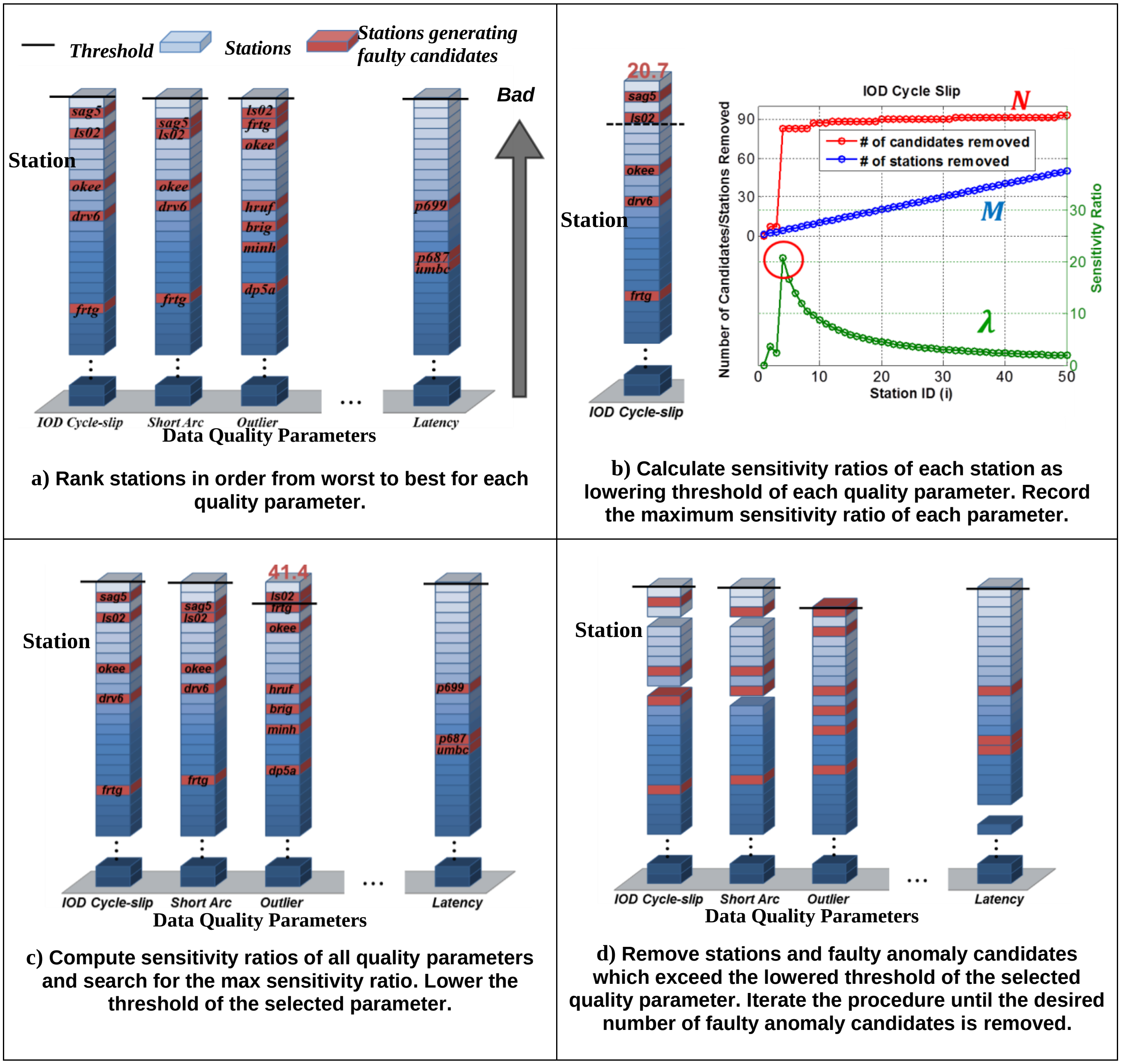
Figure 6-b shows an example result of computing sensitivity ratios of each station as decreasing the threshold of the IOD cycle-slip. The sensitivity ratio, λ, is computed at each ith station (from the worst to the best). When the threshold is lowered to remove the 4th station, LS02, the sensitivity ratio is maximized. Thus, LS02 is the most "desired to be removed" station. Next, to search for the most "efficient" quality parameter to be used, we introduce a two-dimensional search concept. The sensitivity ratios of all stations (1st dimension) for all quality parameters (2nd dimension) are computed, and the quality parameter that returns the maximum sensitivity ratio among all quality parameters is selected. As shown in Figure 6-c, by lowering the threshold of the outlier (instead of the IOD cycle-slip), we can reduce unnecessary removal of nominal stations (from two to one). Then, the threshold of the selected parameter only is lowered at the current step to discard poor-quality stations that exceed the lowered threshold of the selected quality parameter and faulty candidates.

This procedure is iterated until the desired number of faulty anomaly candidates is removed. This method explained above is the two dimensional iterative maximum sensitivity ratio search method. This new method performs better than the previously developed method which conducts a one-dimentional search for stations. The results from processing data in CONUS are shown in Section 4.

3.3. WELL-DISTRIBUTED SUB-NETWORK SELECTION

Stations with poor quality data are determined through the two-dimensional iterative search procedure explained in the previous subsection. However, if a particular station significantly increases the geographical observability of ionospheric behavior in an area, it should be retained despite poor data quality. In contrast, the stations in dense regions do not significantly improve the geographic observability. Thus, these redundant stations, which only bring an excessive computational load, should be eliminated, even if they have high-quality data.

Station recovery processing, which restores stations with poor quality data, has been developed in previous studies. In this method, first, for each station, the coverage of other stations within a 100-km (or a desired baseline distance) radius is examined. "Station coverage" is defined as the area within which pairs of stations whose baseline is less than 100 km form. If a poor-quality station to be removed has another station nearby, the change in station coverage will be small, even after the poor-quality station is removed. However, if the coverage loss after discarding a station is more than 30% of the original coverage, that station is restored despite its poor data quality.



a) Rank stations in order from worst to best for each quality parameter.

b) Calculate sensitivity ratios of each station as lowering threshold of each quality parameter. Record the maximum sensitivity ratio of each parameter.

c) Compute sensitivity ratios of all quality parameters and search for the max sensitivity ratio. Lower the threshold of the selected parameter.

d) Remove stations and faulty anomaly candidates which exceed the lowered threshold of the selected quality parameter. Iterate the procedure until the desired number of faulty anomaly candidates is removed.

Figure 6. The Two-dimensional Iterative Maximum Sensitivity Ratio Search Method to Determine the Thresholds of Quality Parameters

Selecting a subset of stations passing the data quality check (i.e., eliminating redundant stations in dense regions) based upon their geographical distribution is the final step of the optimized GNSS network station selection. This subset selection process combines the two methods, Voronoi method and baseline-distance check, to choose an optimal subset of stations by which an area of interest is sufficiently covered. The Voronoi method, a commonly-used method to divide space into a number of regions [11], is utilized to select a sub-network in which stations are evenly distributed. A station in a dense region has a smaller Voronoi region and, thus, to remove redundant stations, we remove the station with the smallest Voronoi region first (as an example, the dark gray area in Figure 7). After the station is discarded, we redistribute the Voronoi diagram of all of remaining stations and iteratively discard the station with the smallest Voronoi region.

However, the Voroni method does not consider the loss of coverage. Thus, before discarding the station chosen by the Voronoi method, we perform the baseline distance check which is used to maintain an area covered by stations with separations of less than a desired baseline constraint. As shown in Figure 8, if a desired baseline constraint is 40 km, the coverage of stations within a 40-km radius is examined. The station coverage is defined as the area within which pairs of stations whose baseline is less than 40 km form. When the loss of coverage after discarding the station chosen by the Voronoi method is greater than 10%, the station is retained.

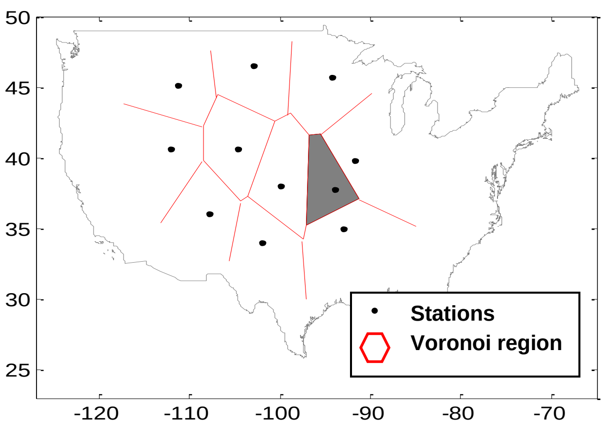


Figure 7. Example of Voronoi Region

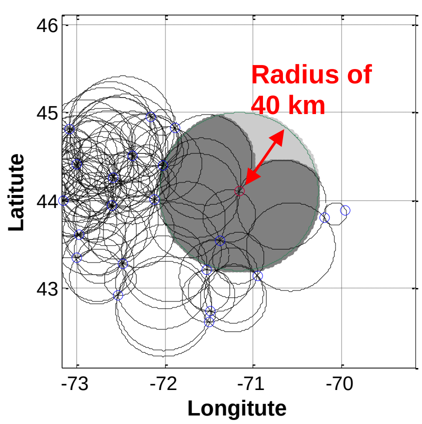


Figure 8. Check coverage by stations with separations of less than 40 km

4.0. RESULTS OF CORS STATION SELECTION

Table 1 shows the dates from which CONUS data were collected and analyzed to evaluate the performance of the optimized GNSS network station selection algorithms. The geomagnetic conditions on these seven consecutive days are shown with two indices of global geomagnetic activity from space weather databases: planetary K (Kp) and disturbance storm time (Dst). In this period, a total of 1587 CORS network stations were operating in CONUS. As Kp and Dst in **오류! 참조 원본을 찾을 수 없습니다.** indicate, the days on which the geomagnetic storm condition was very quiet were chosen to observe CORS station data quality while minimizing any influence of abnormal ionospheric behavior. Since it is known that an anomalous ionospheric event did not occur in this period, the threat candidates resulting from LTIAM processing are known to be faulty candidates generated by processing of poor-quality data.

Table 1. Dates Analyzed to Investigate CORS Network Station Quality in CONUS

Day (UT mm/dd/yy)	Kp	Dst
24/05/12	2.0	-15
25/05/12	2.3	17
26/05/12	2.3	-6
27/05/12	1.3	14
28/05/12	2.3	23
29/05/12	2.3	23
30/05/12	2.3	16

4.1. RESULTS OF HIGH QUALITY STATION SELECTION

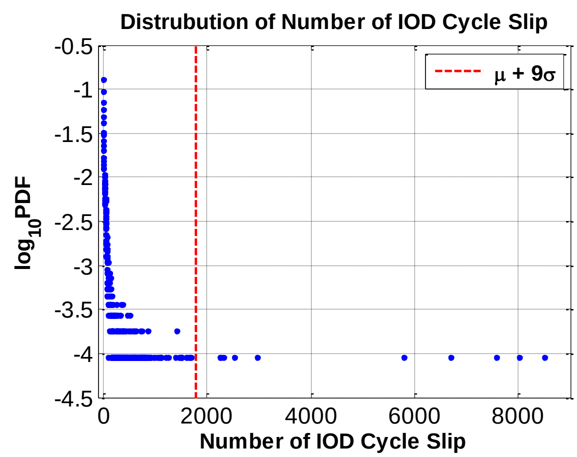


Figure 9. Probability Density Function of the number of IOD Cycle slip for Each Station per Day (data collected for 7 days)

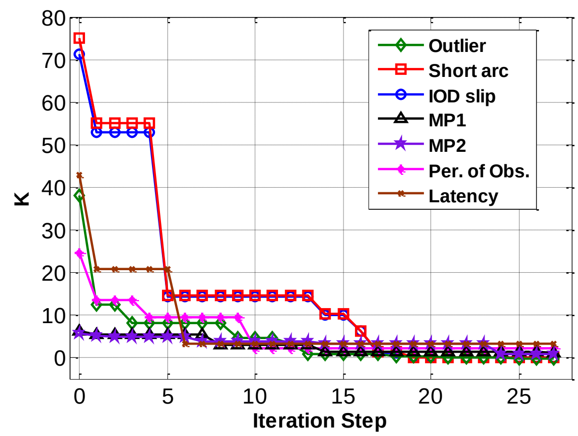


Figure 10. K Values of Seven Quality Parameters at Each Iteration Step

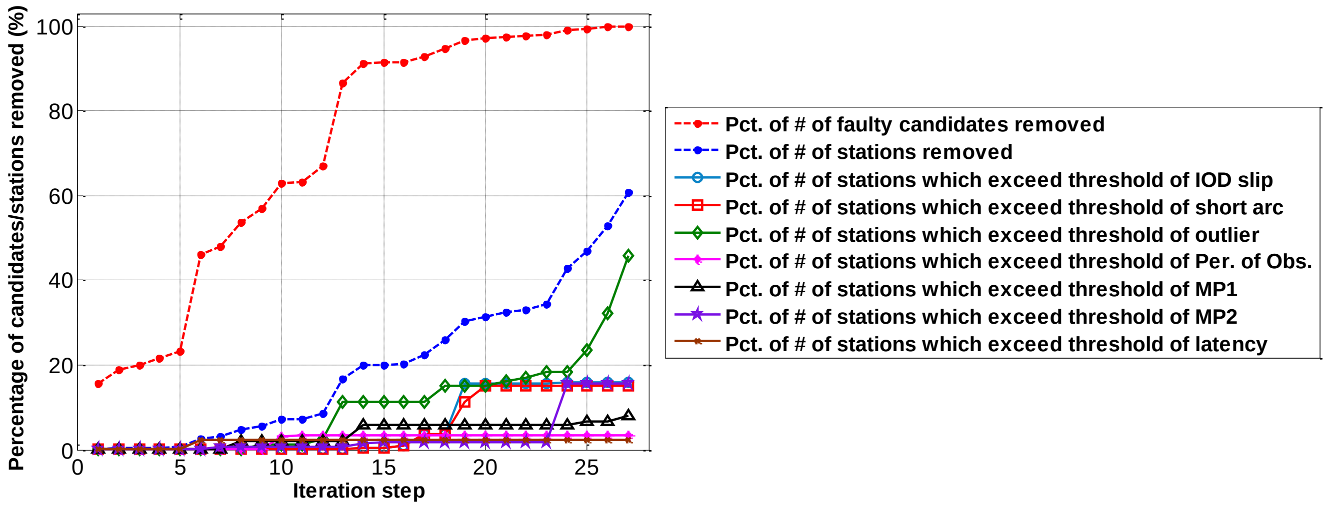


Figure 11. Percentage of the Number of Stations/Candidates Removed at Each Iteration Step

The results from the GNSS data-quality measurement algorithms for the stations and the rank of these stations for each quality parameter are described in [7]. Once the data from CORS stations in CONUS are collected for seven consecutive days (or longer), we can obtain the statistical distribution of each quality parameter. Figure 9 shows the probability density function (PDF) of the number of IOD cycle slips on each station per day in logarithmic scale. The red vertical line in Figure 9 refers to the value of $\mu+9\sigma$ (the mean value plus 9 times the sample standard deviation). Since data (blue) exists continuously from 0 to this line and the continuity of data ceases beyond this line, data that goes beyond $\mu+9\sigma$ are considered to be extreme outliers and are discarded from the distribution.

The threshold of each quality parameter for station selection is expressed as the mean value plus k times sigma ($\mu+k\sigma$) of a revised data distribution that excludes any extreme outliers from the original distribution. The process of high-quality station selection is conducted using the data quality measurements of seven quality metrics. The station and quality parameter which return the maximum sensitivity ratio is determined at each iteration step as illustrated in Figure 6. The thresholds of seven quality metrics are recorded at each iteration step, and the corresponding k values are shown in Figure 10. As the steps proceed, the thresholds and k values decrease and the number of stations removed and the number of faulty candidates removed increase.

Figure 11 shows the percentage of the number of faulty anomaly candidates removed (as a red dashed curve) and the percentage of the number of stations removed (as a blue dashed curve) at each iteration step. The other curves represent the percentage of the number of stations that exceed the threshold of the corresponding quality metric at each iteration step. The percentage of the number of stations that exceed the threshold of the outlier metric (green) is closest to the percentage of the total number of stations removed (dashed blue). This indicates that the number of outliers has the strongest influence on producing faulty anomaly candidates,

LTIAM processing of all stations on seven consecutive days generated 484 faulty candidates (non-existent ionospheric anomalies). As shown in Figure 11, 100% of faulty candidates are removed in the 27th iteration, but at a price of removing 60.7% of the stations. If the desired percentage of the number of faulty candidates removed is 90%, it is achieved in the 14th iteration while removing only 19.9% of the stations. The thresholds of quality parameters in the 14th iteration are shown in Table 2 with the corresponding k values (also see Figure 10). Table 3 shows that the new method performs better that from the one-dimensional one-step search method previously developed. The two-dimensional iterative search method removes more faulty candidates, whereas fewer stations are removed.

Table 2. Thresholds of Data Quality Parameters for CORS Network (the percentage of the number of faulty candidates removed: 91.1%)

Quality parameter	Threshold
Number of outliers	$\mu + 0.85\sigma$
Number of IOD cycle slips	$\mu + 9.87\sigma$
Number of short arcs	$\mu + 10.2\sigma$
Mean of MP1	$\mu + 1.47\sigma$
Mean of MP2	$\mu + 2.04\sigma$
Percentage of Observations	$\mu + 3.17\sigma$
Latency	$\mu + 3.35\sigma$

Table 3. Results of High Quality Station Selection on CORS Network in CONUS (based on LTIAM results from 24-30 May 2012)

	2D iterative search method (Current)	1D one-step search method (Previous)
# of stations removed (out of 1587)	441 (91.1%)	420 (86.8%)
# of faulty candidate removed on seven days (out of 484)	316 (19.9%)	323 (20.4%)

4.2. RESULTS OF WELL-DISTRIBUTED SUB-NETWORK SELECTION

We performed station recovery processing on the 316 stations that were initially classified as bad stations. Among those stations, 57 stations were deemed to be "geographically critical" stations and were not removed. These 57 stations and 259 stations removed due to poor quality data are shown in green and red, respectively, in Figure 12. **오류! 참조 원본을 찾을 수 없습니다.**4 summarizes the results from the station recovery process. After restoring these 57 stations, the number of faulty ionospheric anomaly candidates removed does not change much compared to the results before the station recovery process. The high-quality station selection and station recovery processing removed 89% of the total false anomalies while discarding only about 16% of the total stations.

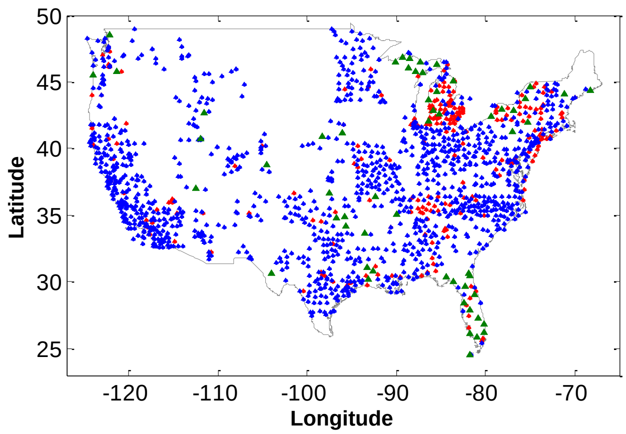


Figure 12. Map of CORS Stations in CONUS (Stations Removed in Red; Stations Restored in Green)

Table 4. Results of Station Recovery Processing on CORS Stations Classified as "Poor Quality" (based on LTIAM results from 24-30 May 2012)

	After high-quality station selection	After station recovery
# of stations removed in CONUS (out of 1587)	316 (19.9%)	259 (16.3%)

# of faulty candidates removed on seven days (out of 484)	441 (91.1%)	432 (89.3%)

Figures 13 and 14 show results from subset selection processing when the baseline constraints are 40 km and 100 km, respectively. Figure 13-a is the Voronoi diagram of the stations that were chosen from the high-quality station selection and station recovery processing. After applying the Voronoi method and baseline distance check as described in Subsection 3.3, a subset of stations are chosen and the Voronoi diagram of the subset stations is shown in Figure 13-b. Because the number of station pairs with baselines of less than 40 km is relatively small, only about 5% of the stations in a dense area were removed by subset selection processing, as shown in Figure 13. However, as illustrated in Figure 14 and Table 5, if a baseline constraint is set to 100 km, the subset selection process reduces the stations down to 46% of the original amount. The stations selected using the proposed method are evenly distributed and the coverage with the desired baseline distance is well maintained.

Table 5. Results of Subset Selection Processing on CORS Stations in CONUS (based on LTIAM results from 24-30 May 2012)

	After station recovery processing	After subset selection processing
# of stations with a Baseline Constraint of 40km (out of 1587)	1328 (83.7%)	1242 (78.3%)
# of stations with a Baseline Constraint of 100km (out of 1587)	1328 (83.7%)	734 (46.3%)

5.0. RESULTS FROM STATION SELECTION OF OTHER GNSS NETWORKS

To validate the performance of the high-quality, well-distributed station selection method, the proposed tool was also tested for other GNSS reference station networks, including Japanese, European, and Canadian GNSS networks. As for a Japanese GPS network, the GPS Earth Observation Network (GEONET) data was used. The EUREF Permanent Network (EPN), integrated geodetic reference network of Germany (GREF), international GNSS service (IGS), and Friuli Regional Deformation Network (FReDNet) are used for European GNSS station selection. The Canadian active control system (CACS) and Canadian high arctic ionospheric network (CHAIN) data were collected to test the algorithms for Canadian GPS networks. The dates from which data were collected and analyzed to test the performance of the optimized GNSS network station selection algorithms are the same dates selected for processing CORS data in Section 4.

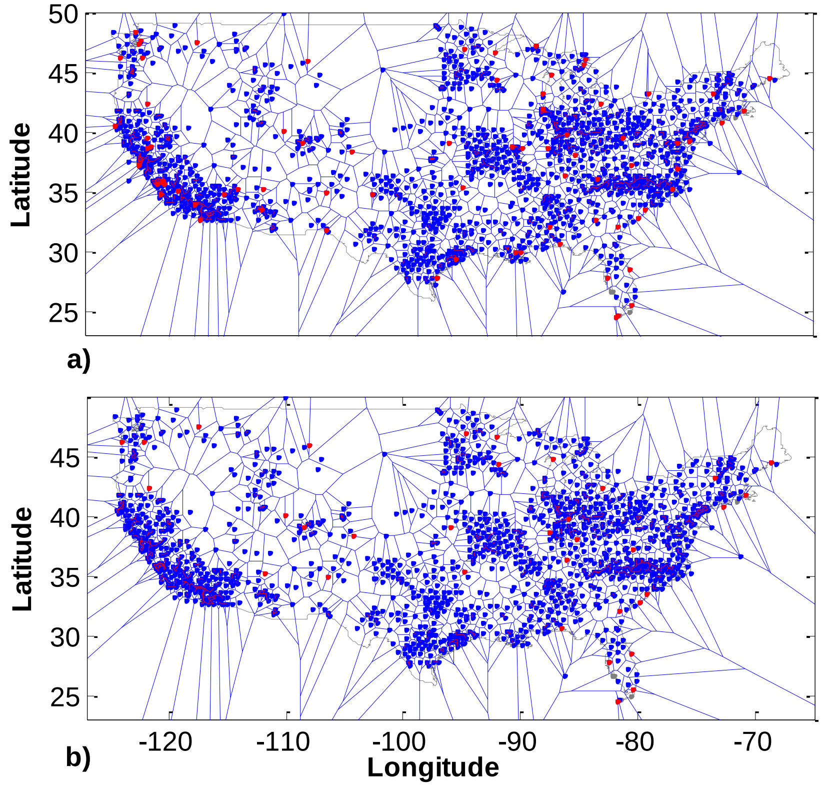


Figure 13. Voronoi Diagram of CORS Stations with Baseline Constraint of 40 km: a) Before Subset Selection Processing b) After Subset Selection Processing (Baselines in Red; Voronoi Regions in Blue Polygon)

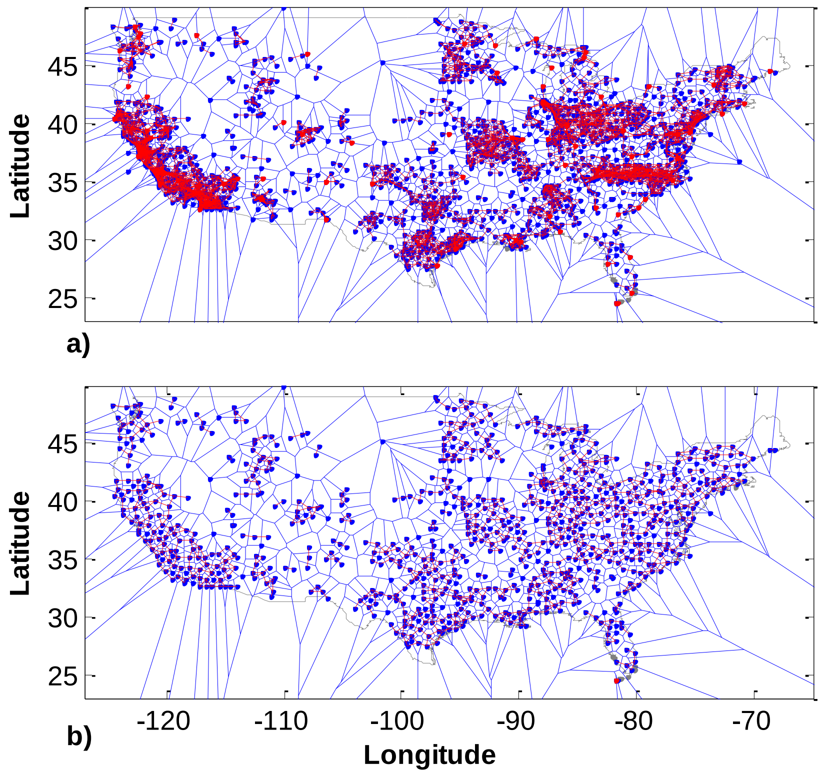


Figure 14. Voronoi Diagram of CORS Stations with Baseline Constraint of 100 km: a) Before Subset Selection Processing b) After Subset Selection Processing (Baselines in Red; Voronoi Regions in Blue Polygon)

In this period, a total of 1229 GEONET stations were operating in Japan. Among those total stations, 19.4% of the total stations were chosen to be removed from the high-quality station selection process. Consequently 90.4% of the total faulty anomaly candidates were discarded, as summarized in **오류! 참조 원본을 찾을 수 없습니다.**6. These results are very similar with those of the CORS network, which implys that the GNSS data qualities of the two networks could be also similar.

Table 6. Results of Station Recovery Processing on GEONET Stations Classified as "Poor Quality" (based on LTIAM results from 24-30 May 2012)

	After high-quality station selection	After station recovery processing
# of stations removed in GEONET (out of 1229)	238 (19.4%)	236 (19.2%)
# of faulty ionospheric anomaly candidates removed on seven days (out of 530)	479 (90.4%)	477 (90.0%)

Figure 15 shows the result from the station recovery process for GEONET. Since GEONET forms a highly dense network in comparison to the CORS network, only two stations were considered to be 'geographically critical' stations among the 238 stations with poor-quality data. Also in the case of GEONET, since many pairs of stations have baselines less than 40 km, as shown in Figure 16 and Table 7, the number of stations could be reduced to approximately 40% of the total through subset selection processing when a baseline constraint is 40 km. Therefore, we need to process only about 40% of GEONET stations and yet maintain the similar level of ionospheric observability.

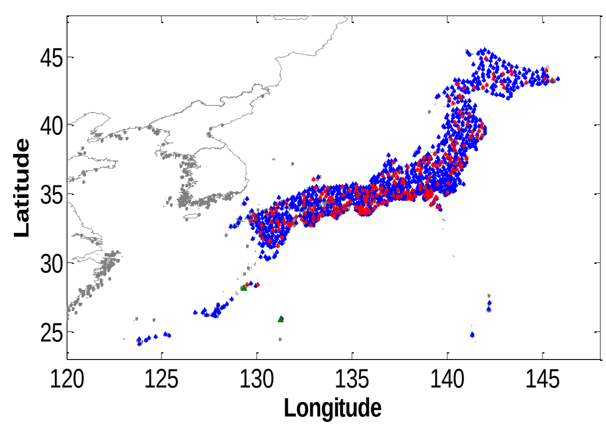


Figure 15. Map of GEONET Stations in Japan (Stations Removed in Red; Stations Restored in Green)

Table 7. Results of Subset Selection Processing on GEONET Stations in CONUS (based on LTIAM results from 24-30 May 2012)

	After station recovery processing	After subset selection processing
# of stations with a Baseline Constraint of 40km (out of 1229)	993 (80.8%)	473 (38.5%)

The results from European and Canadian GPS networks are shown respectively in Figures 17 and 18. In the case of these networks, the tests were performed only up to station recovery processing as the total numbers of stations are relatively small. As summarized in Table 8, for the European GPS networks, 29 (12%) stations out of the total 246 stations were removed and through this approximately 94% of faulty candidates were eliminated. From the test for the Canadian GPS network, 100% of faulty ionospheric anomaly candidates were removed by discarding 7 (16%) stations out of 45 stations, as shown in Table 9. Since the results of other GNSS networks show similar or improved performance compared to the results of the CORS network, it is concluded that the optimized GNSS network station selection method proposed in this study is applicable to a wide range of reference station networks.

Table 8. Results of Station Recovery Processing on European Stations (based on LTIAM results from 24-30 May 2012)

	After station recovery
# of stations removed in European (out of 246)	29 (11.8%)
# of faulty ionospheric anomaly candidates removed on seven days (out of 32)	30 (93.8%)

Table 9. Results of Station Recovery Processing on Canadian Stations (based on LTIAM results from 24-30 May 2012)

	After station recovery
# of stations removed in Canadian (out of 45)	7 (15.6%)
# of faulty ionospheric anomaly candidates removed on seven days (out of 48)	48 (100%)

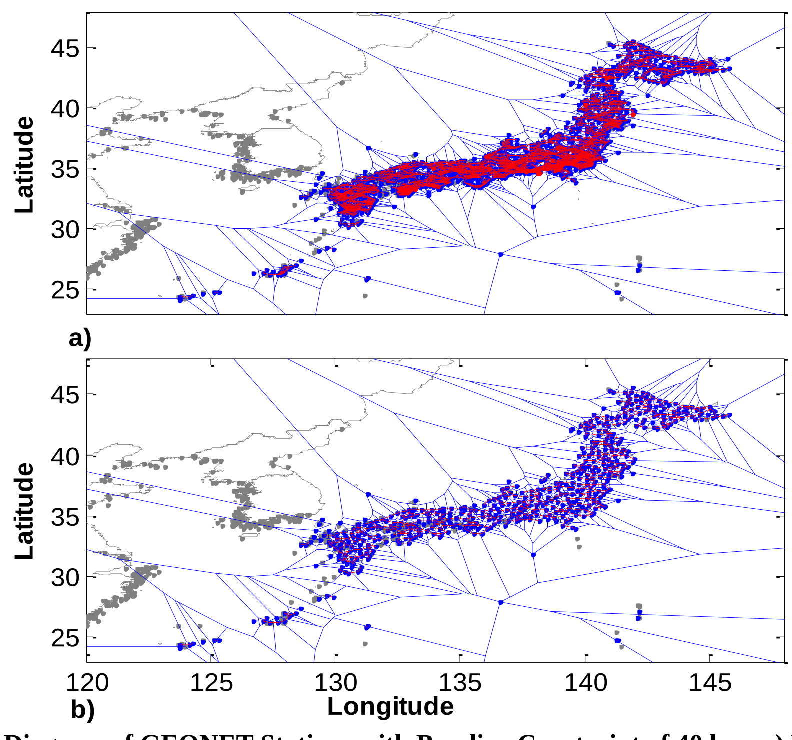


Figure 16. Voronoi Diagram of GEONET Stations with Baseline Constraint of 40 km: a) Before Subset Selection Processing b) After Subset Selection Processing (Baselines in Red; Voronoi Regions in Blue Polygon)

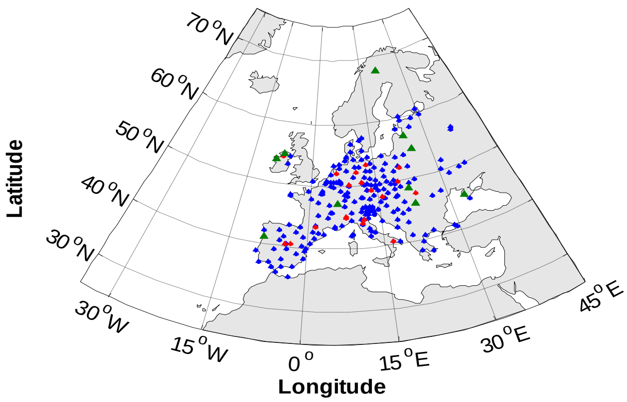


Figure 17. Map of European Stations (Stations Removed in Red; Stations Restored in Green)

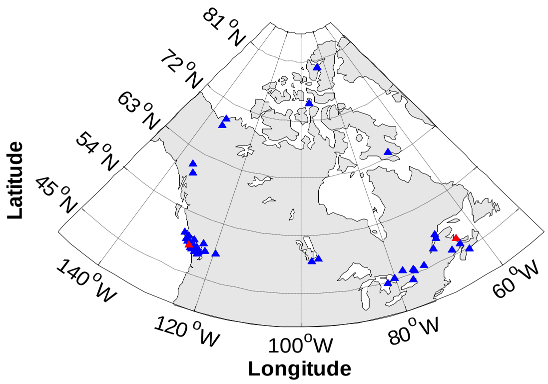


Figure 18. Map of Canadian Stations (Stations Removed in Red; Stations Restored in Green)

6.0. SUMMARY

This paper presents the results of the optimized method to select GNSS network stations with well-distributed, high-quality data and verify the performance of proposed method. Specific stations with poor quality data produce many undesired faulty candidates in LTIAM, and cause excessive processing time in the manual validation. The optimized method developed in this paper achieved a 89% reduction in faulty anomaly candidates while discarding only 16% of CORS stations. The results from processing recently-collected GNSS data confirm the proposed methods also perform well for other GNSS networks. Geographical observability of the ionosphere with a baseline constraint of 40 km can be maintained by processing only 40% of the GEONET stations. Consequently, this method greatly reduces excessive processing time and effort in the manual validation process by faulty candidates and reduces unnecessary processing time resulting from processing redundant station data, where this should benefit the future to building ionosphere threat models for all regions so that GBAS will be fielded in the future.

ACKNOWLEDGMENTS

REFERENCES

[1] Lee, J., Jung, S., Bang, E., Pullen, S., and Enge, P., "Long Term Monitoring of Ionospheric Anomalies to Support the Local Area Augmentation System," *Proceedings of ION GNSS 2010*, Portland, OR, Sept. 21-24, 2010, pp. 2651-2660.

[2] Lee, J., Jung, S., and Pullen, S., "Enhancements of Long Term Ionospheric Anomaly Monitoring for the Ground-Based Augmentation System," *Proceedings of ION ITM 2011*, San Diego, CA, Jan. 24-26, 2011, pp. 930-941.

[3] Lee, J., Jung, S., Kim, M., Seo, J., Pullen, S., and Close, S., "Results from Automated Ionospheric Data Analysis for Ground-Based Augmentation Systems (GBAS)," *Proceedings of ION ITM 2012*, Newport Beach, CA, January 2012, pp. 1451-1461.

[4] Jung. S and Lee, J. "Long-term ionospheric anomaly monitoring for ground based augmentation systems", *Radio science*, Vol. 47, RS4006, July 2012, doi:10.1029/2012RS005016.

[5] Datta-Barua, S., Lee, J., Pullen, S., Luo, M., Ene, A., Qiu, D., Zhang G., and Enge, P., "Ionospheric Threat Parameterization for Local Area GPS-Based Aircraft Landing Systems," AIAA *Journal of Aircraft*, Vol. 47, No. 4, 2010, pp. 1141-1151.

[6] Lee, J., Datta-Barua, S., Zhang, G., Pullen, S., and Enge, P., "Observations of Low-Elevation Ionospheric Anomalies for Ground-Based Augmentation of GNSS,' *Radio Science*, Vol. 46, 2011, RS6005.

[7] Kim, M., Lee, J., Pullen, S., and Gillespie, J., "Data-Quality Improvements and Application of Long Term Monitoring of Ionospheric Anomalies for GBAS", *Proceedings of ION GNSS 2012*, Nashville, TN, Sept. 24-26, 2010.

[8] Larson GPS Research Group, "Bad GPS sites", http://xenon.colorado.edu/reflections/GPS_reflections/BadGPSSites.html

[9] Estey, L. H. and Meertens, C. M., "TEQC: The Multi-Purpose Toolkit for GPS/GLONASS Data", GPS solutions, Vol. 3, Issue 1, 1999, pp. 42-49

[10] UNAVCO, "TEQC Tutorial" , http://facility.unavco.org/software/teqc/tutorial .html

[11] Saalfeld. A, "Voronoi Methods for Spatial Selection", AutoCarto 2012, Columbus, Ohio, Sept. 16-18, 2012